

Evangelis Silva

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Summary

PhD researcher and former senior software engineer with 3+ years of experience designing and optimizing **high-throughput distributed** and **storage systems**.

Specializes in building **distributed storage architectures**, **on-disk data stores**, **database indexing**, **microservices** and **scalable, low-latency data infrastructure** that bridges research innovation with production-grade reliability.

Research Interests: Database and Distributed Systems, Storage Architectures, Indexing & Caching, and Real-time Data Processing

Education

University at Albany, SUNY, Ph.D. in Computer Science Jan 2022 – Present

Advisor: Prof. Jeong Hyon Hwang | Focus: Database and Distributed Systems

University of Moratuwa, Sri Lanka, BS.c. (Hons.) in Information Technology & Management Jan 2014 – Mar 2019

GPA: 3.50/4.20 | Advisor: Dr. Supunmali Ahangama

Experience

Graduate Assistant, University at Albany, SUNY Jan 2022 – Present

- Conduct research on **high-performance on-disk key-value stores** optimized for modern hardware and data-intensive workloads.
- Design and implement **benchmarking frameworks** for throughput, latency, and space-efficiency analysis across diverse workload patterns.
- Assist in teaching undergraduate and graduate computer science courses, supporting labs, discussions, and assessments.

Senior Software Engineer, Sysco LABS, Sri Lanka Nov 2020 – Dec 2021

- Architected and scaled distributed **microservice systems** for Sysco (Fortune 100), supporting **50K+ users** and millions of daily transactions.
- Developed **RESTful APIs** and data pipelines processing over **1TB/day** across regional clusters.
- Improved query performance by **75%**, reducing average latency from 400 ms to 100 ms through indexing and caching optimizations.
- Enabled real-time data synchronization between legacy and microservice databases via **Change Data Capture (CDC)** and **AWS Kinesis**, cutting replication lag to seconds.
- Increased system uptime to **99.98%** through proactive monitoring and distributed tracing (Datadog, Kibana).

Software Engineer, Sysco LABS, Sri Lanka Mar 2019 – Nov 2020

- Developed and deployed **backend services** and **RESTful APIs** powering Sysco's global digital platform for over **20K+ daily users**.
- Implemented **CI/CD pipelines** with Jenkins and Concourse, reducing deployment time by **60%** and improving release reliability.
- Enhanced observability using **Datadog**, **Kibana**, and **Amazon CloudWatch**, reducing mean time to resolution (MTTR) by **40%**.
- Optimized microservice data flows for improved consistency and throughput under **99.9% uptime** SLAs.
- Collaborated with globally distributed teams across the US and Canada to maintain and scale mission-critical systems.

Engineering Intern (Data Analytics), Vesess, Sri Lanka Oct 2017 – Mar 2018

- Developed and automated **data collection and enrichment pipelines** for the *Donordo* analytics platform using Python and web scraping frameworks.
- Built efficient data processing workflows to enhance dataset coverage and preprocessing speed.
- Designed visual analytics dashboards to extract and present **actionable donor-matching insights**.

Technical & Research Skills

Programming, Scripting, and Query Languages: Java, Python, C++, Rust, JavaScript, SQL, Bash

Frameworks & Libraries: Spring Boot, Node.js, AngularJS, NumPy, Pandas, Matplotlib, scikit-learn, NLTK

Cloud & Infrastructure: AWS (EC2, ECS, S3, SNS, SQS, SES, Lambda, EventBridge, Kinesis), Docker, Jenkins, Concourse CI

Databases: PostgreSQL, MongoDB, RocksDB, Amazon DynamoDB, Redis

Data & Systems Tools: Datadog, Kibana, Maven, Gradle, Git, Jira, Elasticsearch

Research & Analysis: Experimental Evaluation, Benchmarking, Data Visualization, System Design & Development, LaTeX

Research & Systems Projects

Mimix Data Transformer (Demo available on YouTube)

- Developed a modular **Java library** for high-throughput event stream transformation, mapping, and post-processing across relational and streaming data systems.
- Designed a **pluggable architecture** supporting dynamic filters, transformation hooks, and custom database writers, packaged as a Maven-compatible artifact.
- Integrated with **AWS Kinesis** for real-time event streaming and replication to downstream systems.
- **Tools and Libraries:** Java, Spring Boot, PostgreSQL, AWS Kinesis, Amazon EC2, Amazon SNS, Amazon S3, Maven, JSON, Jenkins

An Experimental Evaluation of Group Farness Estimation Techniques

- Conducted an extensive empirical study on **scalable group farness estimation** in large-scale graphs, extending single-source algorithms to multi-node group settings.
- Evaluated and benchmarked **sampling-based** and **cardinality estimation** methods (KMV, Cohen's HIP), analyzing accuracy, memory, and runtime trade-offs.
- Demonstrated that Cohen's HIP method achieves **2–5% higher accuracy** and improved scalability across diverse graph structures.
- **Tools and Libraries:** Java, Bash, Python (NumPy, Matplotlib, Pandas)

Enterprise Email Service

- Architected and developed a scalable internal email platform **used by 20+ enterprise services**, enabling automated, template-based communications across Sysco's global products.
- Designed high-throughput RESTful APIs and optimized database schemas, improving message delivery reliability and seamless integration with legacy systems.
- **Tools and Libraries:** Java, Spring Boot, Amazon SNS, Amazon SQS, Amazon EventBridge, Amazon ECS, AWS Lambda, PostgreSQL, Concourse, Jenkins, Datadog, Git, Jira

Compassionate Sri Lanka

- Contributed to the development of a nationwide donor–recipient web platform in collaboration with the Ministry of Social Empowerment, connecting thousands of users across Sri Lanka.
- Developed backend services and RESTful APIs, and built a responsive **Node.js/AngularJS** application with real-time SMS and email notifications for donation tracking and updates.
- **Tools and Libraries:** Node.js, AngularJS, Amazon SES, MongoDB, Git

Publications

E. Silva, M. Nandathilaka, S. Dalugoda, T. Amarasinghe, S. Ahangama, and G. T. Weerasuriya. "Machine Learning-Based Automated Tool to Detect Sinhala Hate Speech in Images." *Proceedings of the 6th International Conference on Information Technology Research (ICITR 2021)*, IEEE.

Awards & Achievements

- **Best Paper Award (AI & ML Track), ICITR 2021 (IEEE)** — For the paper "*Machine Learning-Based Automated Tool to Detect Sinhala Hate Speech in Images.*" (Dec 2021)
- **Customer First Value Award, Sysco LABS (2020)** — For delivering measurable business impact and enhancing customer experience.
- **Dean's List Recognition, University of Moratuwa, Sri Lanka** — Awarded for outstanding academic performance (Level 2, Semester 2).

Professional Service & Leadership

- **Conference Paper Reviewer** — 36th International Conference on Scientific and Statistical Database Management (SSDBM 2024)
- **Student Relations Lead** — Vesess Alpha Coding Challenge (vesess.com/alpha)
- **Treasurer** — Sysco LABS SL Toastmasters Club
- **Editor** — *FIT Chronicles (2017)*, Faculty of IT Newsletter, University of Moratuwa, Sri Lanka
- **Technical Coordinator & Article Reviewer** — *fit2school*, a CSR project of the Faculty of IT, University of Moratuwa, Sri Lanka
- **Organizing Committee Vice President (Delegate)** — IT Recruitment 2016, AIESEC Colombo South